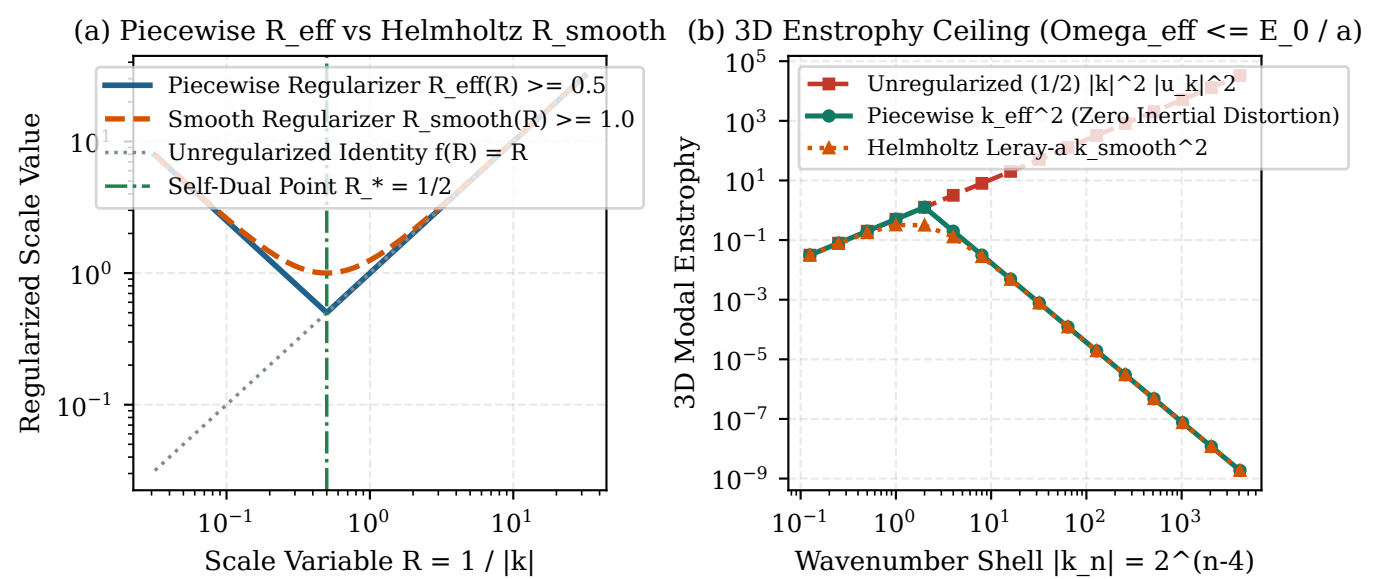
(a) Piecewise R_eff vs Helmholtz R_smooth
Piecewise Regularizer R_eff(R) >= 0.5
Smooth Regularizer R_smooth(R) >= 1.0
Unregularized Identity f(R) = R
Self-Dual Point R_* = 1/2
Regularized Scale Value
Scale Variable R = 1 / |k|
(b) 3D Enstrophy Ceiling (Omega_eff <= E_0 / a)
Unregularized (1/2) |k|^2 |u_k|^2
Piecewise k_eff^2 (Zero Inertial Distortion)
Helmholtz Leray-a k_smooth^2
3D Modal Enstrophy
Wavenumber Shell |k_n| = 2^(n-4)